

Milne's Predictor-Corrector Method

Adams's k step explicit method $\rightarrow$

$$y_{n+1} = y_{n-4} + \sum_{m=0}^{k-1} \beta_m^{(k)} \nabla^m f_n$$

$$\text{where } \beta_m^{(k)} = \int_{-1}^1 (-1)^m \binom{k-s}{m} ds = \int_{-1}^1 \frac{s(s+1)\dots(s+m-1)}{m!} ds$$

Taking $k=3, k=4$, we get

$$y_{n+1} = y_{n-3} + \sum_{m=0}^3 \beta_m^{(3)} \nabla^m f_n$$

$$\beta_m^{(3)} = \int_{-3}^1 \frac{s(s+1)\dots(s+m-1)}{m!} ds \quad m=0,1,2,3$$

$$\beta_0^{(3)} = \int_{-3}^1 ds = 4$$

$$\beta_1^{(3)} = \int_{-3}^1 s ds = \frac{1-9}{2} = -4$$

$$\beta_2^{(3)} = \int_{-3}^1 \frac{s(s+1)}{2} ds = \int_{-3}^1 \frac{s^2+s}{2} ds = \left(\frac{1+27}{6} + \frac{1-9}{4} \right) = \frac{28}{3} - 2 = \frac{8}{3}$$

$$\beta_3^{(3)} = \int_{-3}^1 \frac{s(s+1)(s+2)}{6} ds = \int_{-3}^1 \frac{s^3+3s^2+2s}{6} ds = \frac{1-81}{24} + \frac{1+27}{6} + \frac{1-9}{6} = -\frac{80}{24} + \frac{28}{6} - \frac{8}{6} = 0$$

$$\begin{array}{c|c|c|c|c} m & 0 & 1 & 2 & 3 \\ \hline \beta_m^{(3)} & 4 & -4 & \frac{8}{3} & 0 \end{array}$$

$$y_{n+1} = y_{n-3} + h \left[4f_n - 4\nabla f_n + \frac{8}{3} \nabla^2 f_n \right]$$

$$= y_{n+1} + h \left[4f_n - 4(f_n - f_{n-1}) + \frac{8}{3}(f_n - 2f_{n-1} + f_{n-2}) \right]$$

$$= y_{n+1} + h \left[\frac{8}{3} f_n - \frac{4}{3} f_{n-1} + \frac{8}{3} f_{n-2} \right]$$

$$y_{n+1} = y_{n-3} + \frac{4}{3} h \left[2f_n - f_{n-1} + 2f_{n-2} \right]$$

This is called Milne's Predictor formula. It is of order 4.

Milne's corrector formula $\rightarrow$ Adams's k step implicit formula

is given by

$$y_{n+1} = y_{n-4} + \sum_{m=0}^k \gamma_m^{(k)} \nabla^m f_{n+1}$$

$$\text{where } \gamma_m^{(k)} = \int_{-1}^1 (-1)^m \binom{k-s}{m} ds = \int_{-1}^1 \frac{(s-1)s\dots(s+m-2)}{m!} ds$$

Take $k=1, k=3$

$$y_{n+1} = y_{n-1} + h \sum_{m=0}^3 \gamma_m^{(3)} \nabla^m f_{n+1}, \text{ where}$$

$$\gamma_m^{(3)} = \int_{-1}^1 \frac{(s-1)s\dots(s+m-2)}{m!} ds \quad m=0,1,2,3$$

$$\gamma_0^{(3)} = \int_{-1}^1 ds = 2$$

$$\gamma_1^{(3)} = \int_{-1}^1 (s-1) ds = \left(\frac{s^2}{2} - s \right) \Big|_{-1}^1 = -2$$

$$\gamma_2^{(3)} = \int_{-1}^1 \frac{(s^2-s)}{2} ds = \frac{1}{3}$$

$$\gamma_3^{(3)} = 0$$

$$\begin{array}{c|c|c|c|c} m & 0 & 1 & 2 & 3 \\ \hline \gamma_m^{(3)} & 2 & -2 & \frac{1}{3} & 0 \end{array}$$

$$y_{n+1} = y_{n-1} + \frac{h}{3} [f_{n+1} + 4f_n + f_{n-1}]$$

This is called Milne's corrector formula. It is of order 4.

The Milne's P-C pair (Predictor-corrector pair) is

$$P: y_{n+1} = y_{n-3} + \frac{4}{3} h [2f_n - f_{n-1} + 2f_{n-2}]$$

$$C: y_{n+1} = y_{n-1} + \frac{h}{3} [f_{n+1} + 4f_n + f_{n-1}]$$

(looks like Simpson's one-third rule)

Predictor-corrector schemes $\rightarrow$

① PECE scheme - This is explained as follows $\rightarrow$

i) P - Find an approx. for y_{n+1} using the predictor formula

and denote it by $y_{n+1}^{(P)}$.

ii) E - Evaluate $f(x_{n+1}, y_{n+1}^{(P)}) = f_{n+1}^{(P)}$

iii) C - Find the new approx. for y_{n+1} by using $f_{n+1}^{(P)}$

$= f(x_{n+1}, y_{n+1}^{(P)})$ on RHS of corrector formula.

Denote it by $y_{n+1}^{(C)}$.

iv) Evaluate $f_{n+1}^{(C)} = f(x_{n+1}, y_{n+1}^{(C)})$

We then go to the next step / iteration / nodal point.

Solution at x_{n+2} is

$$P: y_{n+2}^{(P)} = y_{n-2} + \frac{4}{3} h [2f_{n+1}^{(C)} - f_n + 2f_{n-1}]$$

$$E: f_{n+2}^{(P)} = f(x_{n+2}, y_{n+2}^{(P)})$$

$$C: y_{n+2}^{(C)} = y_{n-2} + \frac{h}{3} [f_{n+2}^{(P)} + 4f_{n+1}^{(C)} + f_n]$$

$$E: f_{n+2}^{(C)} = f(x_{n+2}, y_{n+2}^{(C)})$$

Then go to the next step.

② P(EC)²E scheme

PEC²EC²E

In this, for the next iteration, the evaluated value of f_{n+1} using $y_{n+1}^{(P)}$ is used, instead of the one using $y_{n+1}^{(C)}$ like in PECE.

③ PEC scheme

We also have PEC²EC = P(EC)² scheme.

Simplest predictor-corrector method

$$P: y_{n+1} = y_n + h f(x_n, y_n) \quad 1^{st} \text{ order}$$

$$C: y_{n+1} = y_n + \frac{h}{2} [f_{n+1} + f_n] \quad 2^{nd} \text{ order}$$

PECE $\rightarrow$

$$P: y_1^{(P)} = y_0 + h f(x_0, y_0)$$

$$E: f_1^{(P)} = f(x_1, y_1^{(P)})$$

$$C: y_1^{(C)} = y_0 + \frac{h}{2} [f_1^{(P)} + f_0]$$

$$E: f_1^{(C)} = f(x_1, y_1^{(C)})$$

Proceed to next step.

Problem Use Milne's predictor-corrector method to obtain the solution of $y' = f(x, y)$ at $x = 1.2$ and $x = 1.5$ given that $y(0) = 0, y(0.3) = 0.0496, y(0.6) = 0.221, y(0.9) = 0.5596$ using PECE scheme.

(If nothing is mentioned, PECE is default scheme).

+ If initial values not given, use single step method to find them.

$$\underline{\text{Solution}} \quad P: y_{n+1} = y_{n-3} + \frac{4}{3} h [2f_n - f_{n-1} + 2f_{n-2}]$$

$$C: y_{n+1} = y_{n-1} + \frac{h}{3} [f_{n+1} + 4f_n + f_{n-1}]$$

$$\underline{n=3} \quad P: y_4 = y_0 + \frac{4h}{3} [2f_3 - f_2 + 2f_1]$$

$$C: y_4 = y_2 + \frac{h}{3} [f_4 + 4f_3 + f_2]$$

$$f(x, y) = x + y \quad [\text{given}]$$

$$f_1 = f(x_1, y_1) = x_1 + y_1 = 0.3 + 0.0496 = 0.3496$$

$$f_2 = 0.8221$$

$$f_3 = 1.4596$$

$$P: y_4^{(P)} = 0 + \frac{4}{3} \times 0.3 [2 \times 1.4596 - 0.8221 + 2 \times 0.3496] = 1.1187$$

$$E: f_4^{(P)} = f(x_4, y_4^{(P)}) = 1.2 + 1.1187 = 2.3187$$

$$C: y_4^{(C)} = 0.2221 + \frac{0.3}{3} [2.3187 + 4 \times 1.4596 + 0.8221] = 1.1200$$

$$E: f_4^{(C)} = f(x_4, y_4^{(C)}) = 1.2 + 1.1200 = 2.3200$$

$$y(1.2) \sim y_4^{(C)} = 2.3200$$

Now, for $n=4 \rightarrow$

$$P: y_5 = y_1 + \frac{4h}{3} [2f_4^{(C)} - f_3 + 2f_2]$$

$$C: y_5^{(C)} = y_3 + \frac{h}{3} [f_5^{(P)} + 4f_4^{(C)} + f_3]$$

$$f_2 = 0.8221, \quad f_3 = 1.4596, \quad f_4^{(C)} = f_4 = 2.3200$$

$$P: y_5^{(P)} = 0.04967 + \frac{4 \times 0.3}{3} [2 \times 2.3200 - 1.4596 + 2 \times 0.8221] = 1.9798$$

$$E: f_5^{(P)} = f(x_5, y_5^{(P)}) = 1.5 + 1.9798 = 3.4798$$

$$C: y_5^{(C)} = 0.5596 + \frac{0.3}{3} [3.4798 + 2.32 \times 2 + 1.4596] = 1.9816$$

$$E: f_5^{(C)} = 1.5 + 1.9816 = 3.4816$$

x	0	1	2	3	4	5
y	0	0.2	0.6	0.9	1.2	1.5
f	0	0.496	0.8221	0.5596	1.1200	1.9816

$$y' = f(x, y) \quad y(x_0) = y_0$$

$$\int_{x_{n-1}}^{x_{n+1}} y' dx = \int_{x_{n-1}}^{x_{n+1}} \frac{f(x, y(x))}{\text{Approximated by Simpson's } \frac{1}{3} \text{ rule}} dx$$

$$y(x_{n+1}) - y(x_{n-1}) \approx \frac{h}{3} [f(x_{n-1}, y_{n-1}) + 4f(x_n, y_n) + f(x_{n+1}, y_{n+1})]$$

$$\rightarrow y_{n+1} = y_{n-1} + \frac{h}{3} [f_{n-1} + 4f_n + f_{n+1}]$$

This is Milne's corrector method (a 2 step method).

General linear k step method

$$\sum_{j=0}^k \alpha_j y_{n-j} = h \sum_{j=0}^k \beta_j f(x_{n-j}, y_{n-j}) \quad \text{--- (1)}$$

$\alpha_0, \alpha_1, \dots, \alpha_k$ and $\beta_0, \beta_1, \dots, \beta_k$ are real constants

such that $\alpha_0^2 + \beta_0^2 \neq 0 \Leftrightarrow |\alpha_0| + |\beta_0| = 0$ [is k step]

[$\beta_k = 0 \rightarrow$ explicit method, $\beta_k \neq 0 \rightarrow$ implicit method]

Why linear? Uses only linear combinations of $\{y_n\}, \{f(x_n, y_n)\}$

Ex 4 step Adams-Bashforth method

$$y_{n+4} = y_{n+3} + \frac{1}{24} h [55f_{n+3} - 59f_{n+2} + 37f_{n+1} - 9f_n]$$

is an example of an explicit 4 step general linear method.

4 step Adams-Moulton method

$$y_{n+4} = y_{n+3} + \frac{1}{24} h [9f_{n+4} + 19f_{n+3} - 5f_{n+2} - 9f_{n+1}]$$

is an eq. of implicit 4 step method.

Zero stability

A linear k step method is said to be zero stable if $\exists$ a const. k, ϵ , for any 2 seq. $\{y_n\}, \{\tilde{y}_n\}$ which have been generated by the same formula but diff. initial data $y_0, y_1, \dots, y_{n-1}$ and $\tilde{y}_0, \tilde{y}_1, \dots, \tilde{y}_{n-1}$ respectively, we have

$$|y_n - \tilde{y}_n| \leq k \max \{ |y_0 - \tilde{y}_0|, |y_1 - \tilde{y}_1|, \dots, |y_{n-1} - \tilde{y}_{n-1}| \}$$

A general k step method in (1), consider its 1st & 2nd characteristic polynomial respectively.

$$S(z) = \sum_{j=0}^k \alpha_j z^j$$

$$C(z) = \sum_{j=0}^k \beta_j z^j \quad \text{where } \alpha_k \neq 0, \alpha_0^2 + \beta_0^2 \neq 0$$

A linear multistep method is zero stable for any ODE of the form $y' = f(x, y)$ where f satisfies the Lipschitz condition $|f(x, y) - f(x, z)| \leq L|y - z|$ iff its 1st characteristic polynomial

Roots of 1st characteristic polynomial lie in the closed unit disc & those on the circle are simple, is often called root condition.